

Introduction to the Theory of Formal Languages

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Formal Language Theory

Finite representations of infinite sets of objects

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Finite representations of infinite sets of objects

words

a — a — b — c — ...

Formal Language Theory

Finite representations of infinite sets of objects

discrete
behavior

a — a — b — c — ...



Formal Language Theory

Finite representations of infinite sets of objects

real-time
behavior

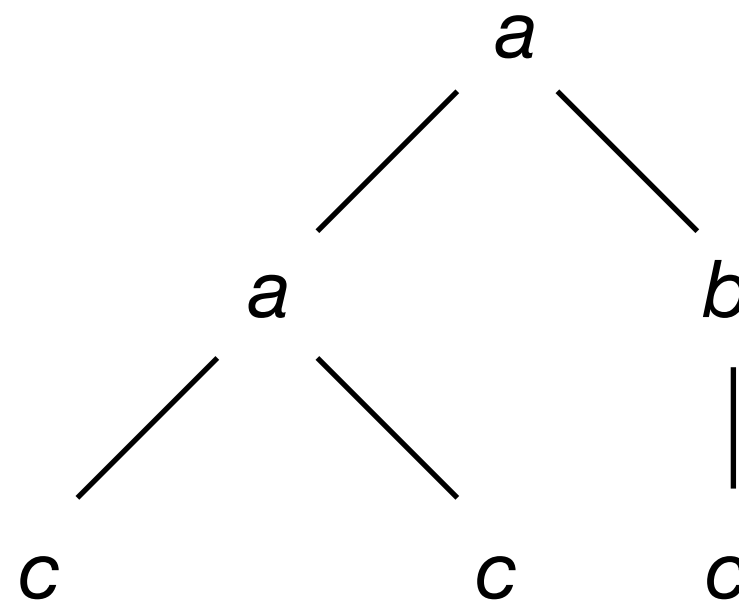
$$a \xrightarrow{0.1} a \xrightarrow{0.5} b \xrightarrow{0.3} c \xrightarrow{0.4} \dots$$



Formal Language Theory

Finite representations of infinite sets of objects

trees



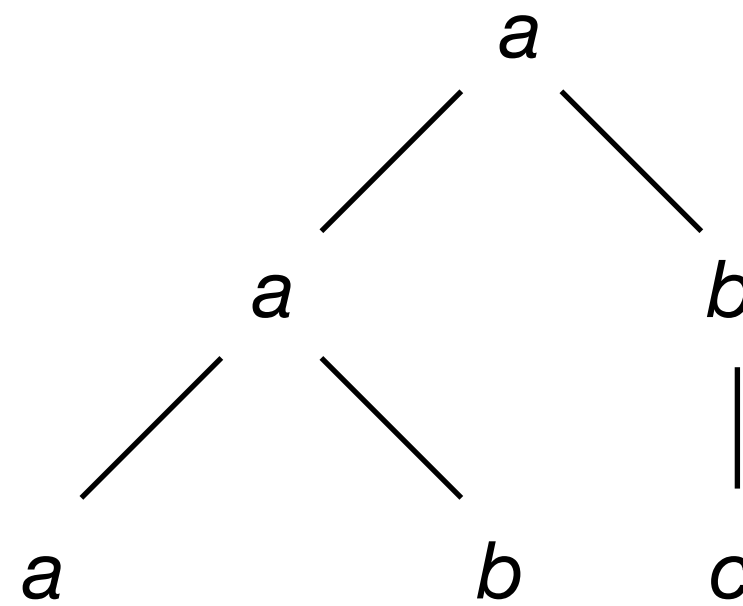
terms

$a(a(c,c),b(c))$

Formal Language Theory

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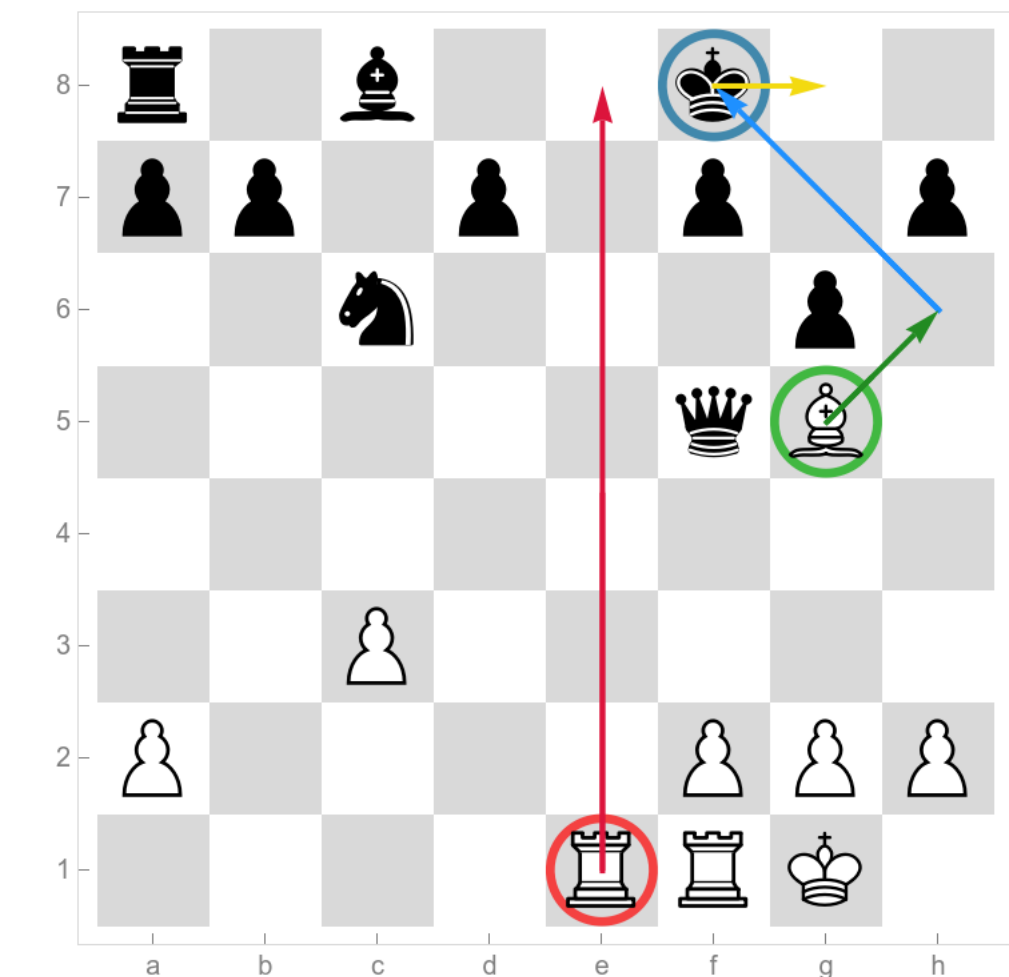
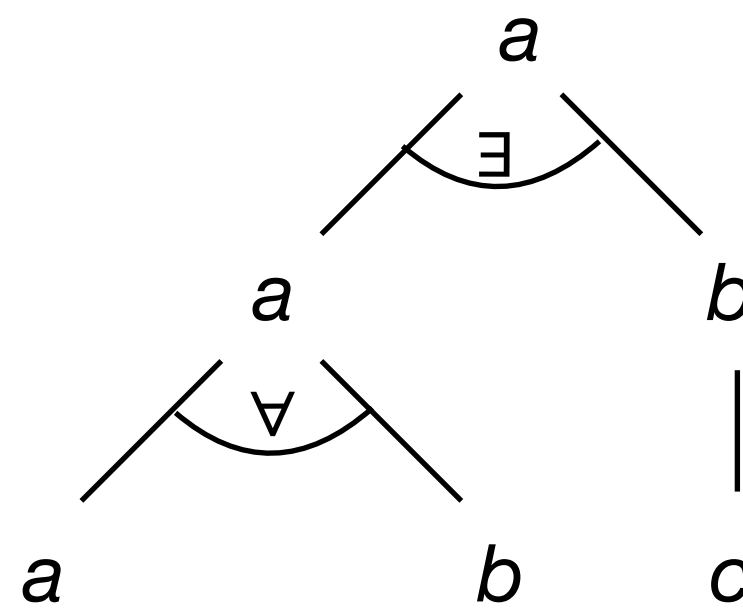
non-deterministic
behavior



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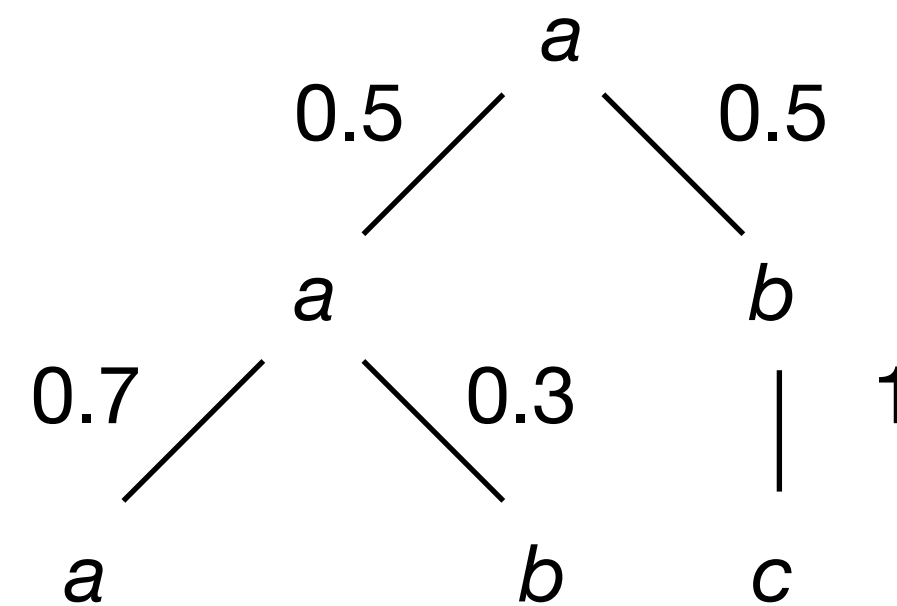
game-theoretic
behavior



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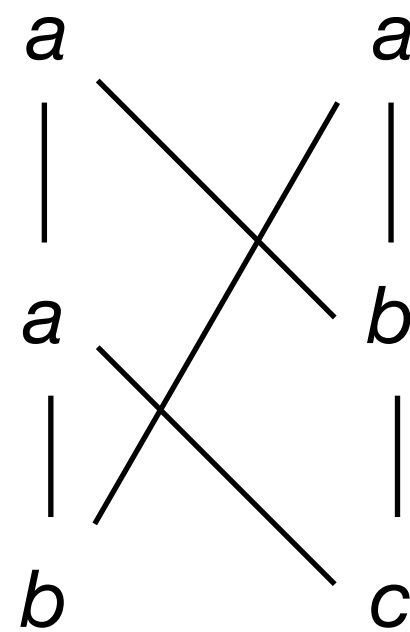
stochastic
behavior



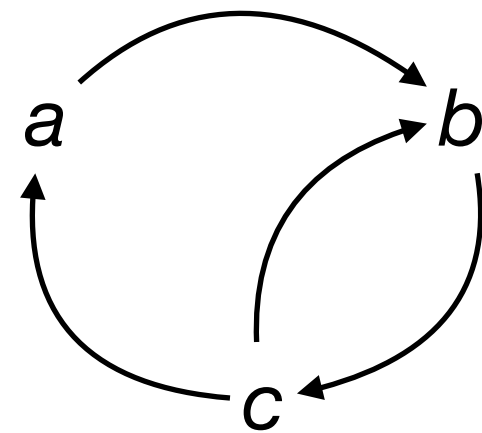
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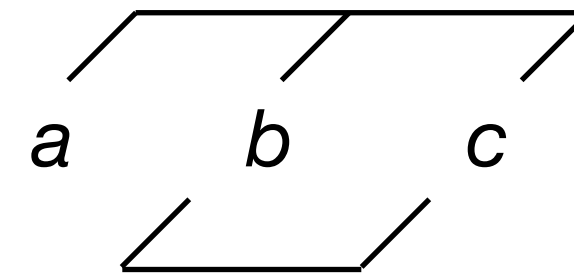
graphs



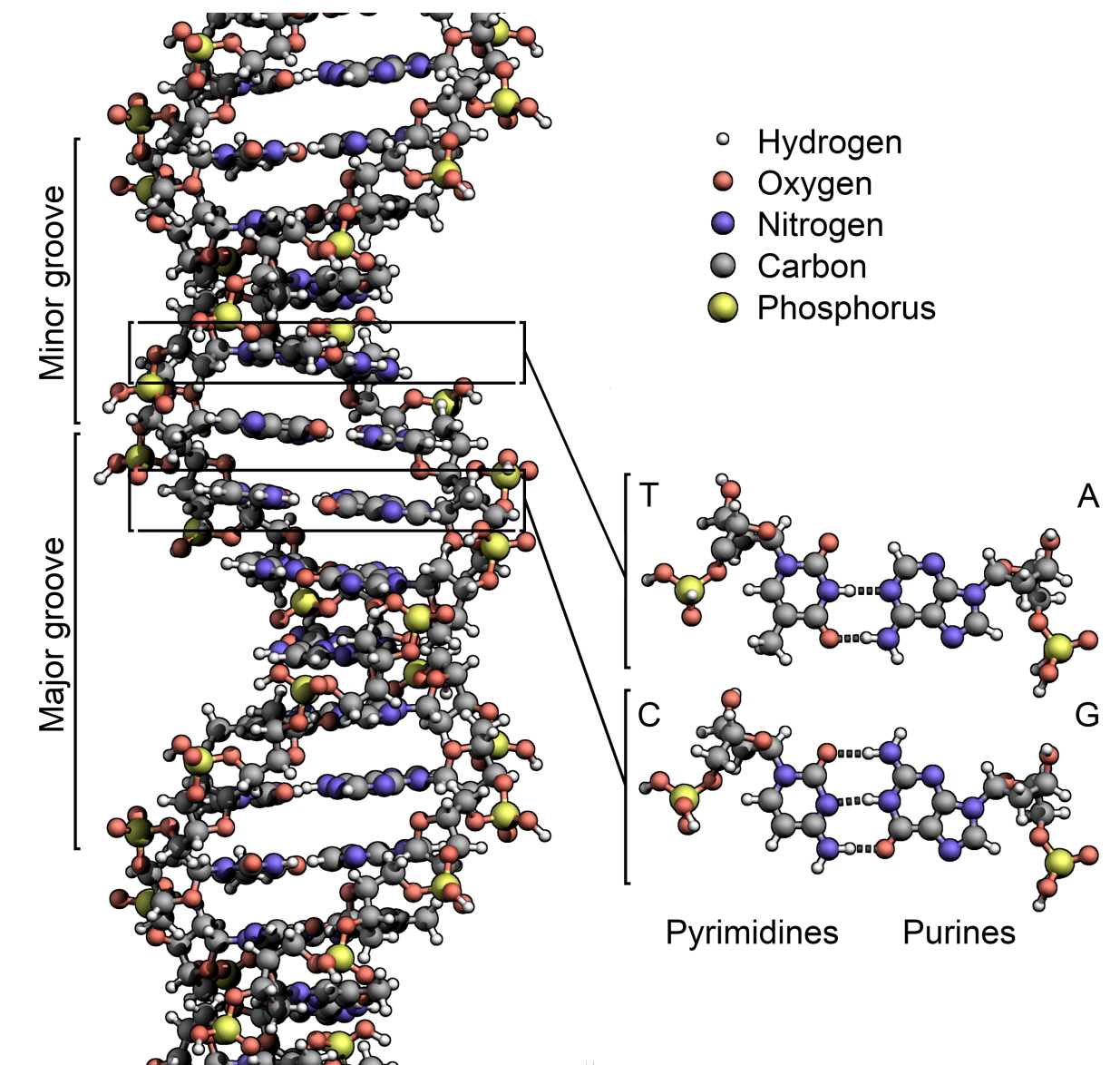
message
sequence
charts



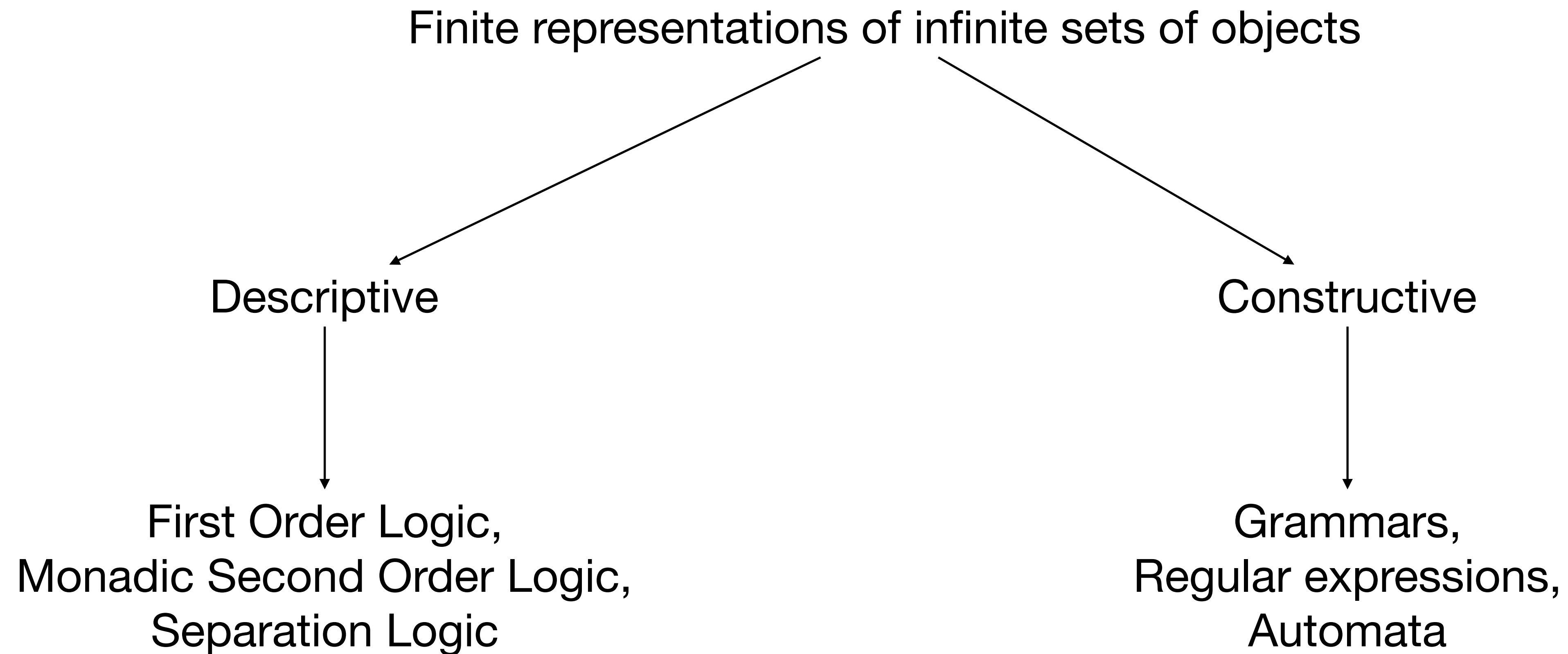
data structures,
networks, ...



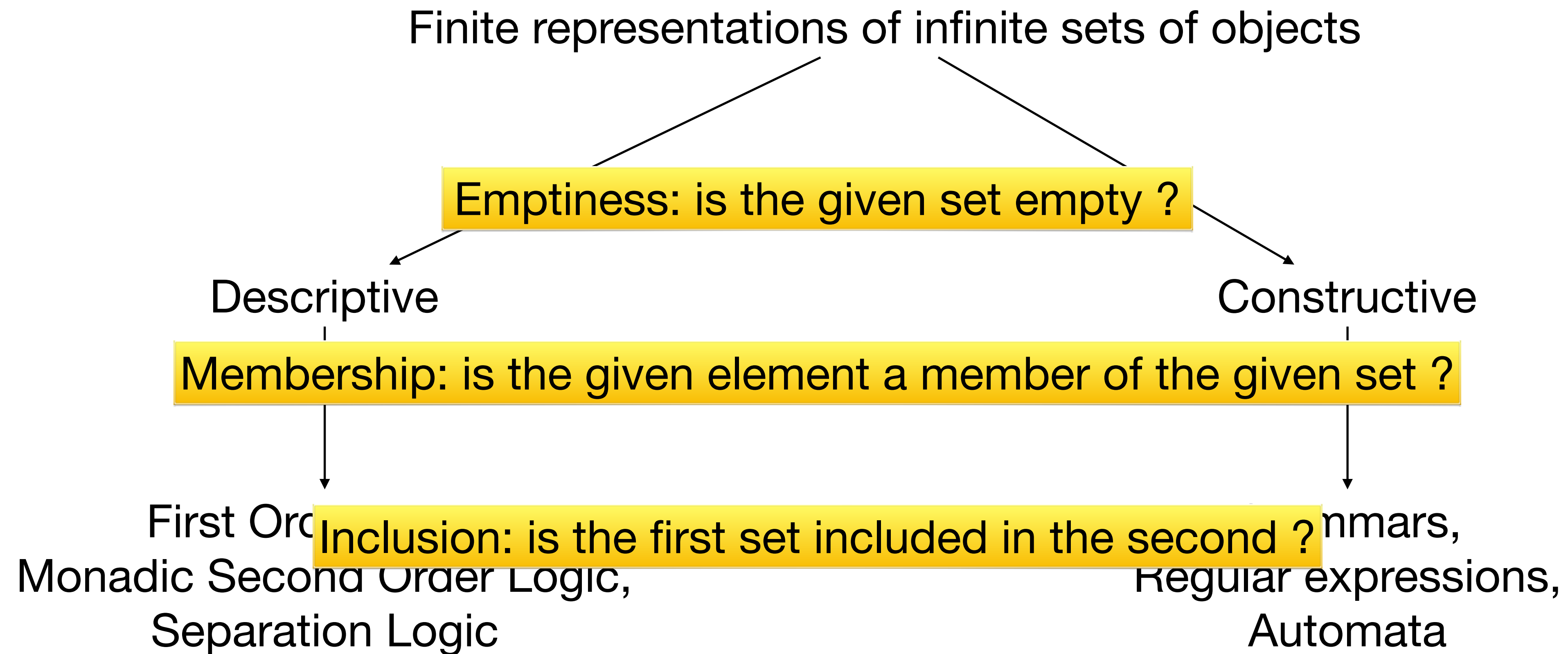
relational
databases



Types of Representation



Types of Representation



Inclusions and Equivalences

words

MSO
=
recognisable
(word automata)
⊆
context-free
(grammars)

terms

MSO
=
recognisable
(tree automata)
=
context-free
(grammars)

graphs

CMSO
⊊
recognisable
(locally-finite algebras)
≠
context-free
(grammars)

Inclusions and Equivalences

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 $\subseteq$
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emptiness,
membership
inclusion

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Outline

- words {
 - 1. Recognisability by semigroups and monoids
 - Definability of sets recognisable by monoids (Monadic Second Order Logic)
 - Aperiodic monoids (First Order Logic)
- trees {
 - 2. Recognisability by algebras
 - Context-free (equational) sets
- graphs {
 - 3. Graph algebras (Hyperedge Replacement)
 - Recognisability of MSO-definable sets of graphs
 - Logical characterisation of context-free sets of graphs